

- Q1. Define shock and describe pathophysiology of shock.**
- Q2. Classify and describe the signs, symptoms and management of shock.**
- Q3. Discuss the different types of haemorrhage and their management.**
- Q4. Describe the pathophysiology, clinical features and treatment of septic shock.**
- Q5. Describe pathophysiology, clinical features of hemorrhagic shock . How will you manage a case of primary hemorrhage after a dental extraction.**
- Q6. Describe neurogenic shock and its management.**
- Q7. Describe indications and complications of blood transfusion.**
- Q8. Discuss the Management of Hypovolaemic Shock.**
- Q9. Triage trauma**
- Q10. Causes , treatment and management of vasovagal shock**
- Q1. Define shock and describe pathophysiology of shock.**

Shock

Definition:

Shock is an **acute clinical syndrome** caused by a **systemic reduction in tissue perfusion (hypoperfusion)**, leading to:

1. Decreased **tissue oxygen delivery**
2. Insufficient removal of **metabolic waste**
3. Resulting in **tissue injury** and **organ dysfunction**

Pathophysiology of Shock

Two basic processes:

1. **Decrease in effective circulating volume:**
 - **Actual loss** of blood volume (e.g., hemorrhage).
 - **Decreased cardiac output** without volume loss (normovolemia).

2. Hypoxia/Anoxia of tissues:

- **Low venous return** → ↓ cardiac output → ↓ oxygen supply.
- Results in **tissue hypoxia** and organ dysfunction.

Pathogenesis

1. Cellular Ischemia:

- **Hypoperfusion** causes:
 - **Low ATP production** → affects energy-dependent processes.
 - **Lysosomal enzyme damage** → irreversible injury.
 - Accumulation of **lactic acid** → **metabolic acidosis**.

Flowchart:

- ↓ ATP → **Failure of Na/K pump** → **Na⁺ and water influx** → cell swelling.
 - ↓ Oxygen → **Anaerobic metabolism** → **Lactic acid build-up**.
 - **Vasoconstriction** → capillary damage → further **cell death**.
- ### 2. Inflammatory Mediators:
- Seen in **septic shock** or **trauma-related hemorrhagic shock**.
 - **Endotoxin** (gram-negative bacteria) triggers inflammation.
 - **Mediators**:
 - **Cytokines**: TNF-α, IL-1β.
 - Others: IL-2, IL-6, prostaglandins, complement fragments (C3a, C5a).
- ### 3. Free Radical Injury:
- **Ischemia** → build-up of **adenosine, inosine, hypoxanthine**.
 - During **reperfusion** (O₂ supply):
 - Formation of **superoxide (O₂⁻)** and **hydrogen peroxide (H₂O₂)**.
 - Damage from **hydroxyl radicals** → attracts neutrophils → amplifies injury.

Stages of Shock

1. Non-progressive (Compensated, Reversible) Shock

- **Initial stage** where **hypoperfusion** leads to:
 - **Hypoxia** → Mitochondria fail to produce **ATP**.
 - **Cell damage**: Membranes become **leaky**, and cells perform **anaerobic respiration**.
 - Build-up of **lactic acid** and **pyruvic acid** → **Metabolic acidosis**.
 - **Liver cannot clear acids** due to lack of oxygen.
 - **Redistribution of blood** maintains supply to vital organs like the **brain and heart**.

2. Progressive (Decompensated) Shock

- **Body compensates for poor tissue perfusion** using physiological mechanisms:
 - **i. Adrenergic discharge:**
 - Vasoconstriction of **splanchnic viscera, skin,** and kidney vessels.
 - Improves **right heart filling**, increases **cardiac output**, and diverts blood to **brain and heart**.
 - **ii. Hyperventilation:**
 - Deep breathing compensates for **metabolic acidosis**.
 - Improves **venous return** → increases **left ventricle filling** and **stroke volume**.
 - **iii. Vasoconstrictive hormones:**
 - Activation of **Renin-Angiotensin-Aldosterone System (RAAS)**, ADH, and epinephrine.
 - **iv. Resorption of fluid from interstitial space:**
 - Vasoconstriction reduces **intravascular pressure**, pulling **water and electrolytes** into capillaries.
 - **v. Intracellular to extracellular fluid shift.**
 - **vi. Renal water conservation** → body retains **water and electrolytes**.

3. Irreversible (Decompensated) Shock

- Stage of **no return**, characterized by:
 - **Persistent vasoconstriction** and increased **vascular permeability**.
 - **Severe hypoxia** → **Cell death** → leads to **MODS (Multiple Organ Dysfunction Syndrome)**.

Q2. Classify and describe the signs, symptoms and management of shock.

Classification of Shock

(Classification by **Hinshaw and Cox, 1972**)

1. Hypovolemic Shock:

- **Cause:** Loss of blood, plasma, or body fluids (e.g., **massive hemorrhage**, vomiting, diarrhea, dehydration).
- **Mechanism:**
 - ↓ **Circulatory volume** → ↓ **venous return** → ↓ **cardiac output** → ↑ **SVR (Systemic Vascular Resistance)**.
- **Most common type** of shock.

2. Cardiogenic Shock:

- **Cause:** Dysfunction of heart ventricles (e.g., **myocardial infarction (MI)**, cardiac arrhythmias, pulmonary embolism).
- **Mechanism:**
 - Heart fails to pump blood adequately → ↓ **cardiac output**.
- **MI is the most common cause.**

3. Distributive Shock:

- **Cause:** Excessive **vasodilation** and maldistribution of blood flow.
- **Types:**
 - **Septic Shock:**
 - Due to **endotoxins** from Gram-negative bacteria (e.g., **E. coli**, **Klebsiella**) → widespread vasodilation.
 - **Neurogenic Shock:**
 - Loss of **sympathetic tone** (e.g., spinal injury, spinal anesthesia, drugs).
 - **Anaphylactic Shock:**
 - Severe **allergic reaction (Type I hypersensitivity)** due to triggers like **penicillin, contrast media, serum**.

4. Obstructive Shock:

- **Cause:** Physical obstruction of **great vessels** or **heart** (e.g., **pulmonary embolism, cardiac tamponade**).
- Similar to **cardiogenic shock**.

Signs and Symptoms of Shock

1. **CNS Symptoms:**
 - **Anxiety, restlessness, altered mental state** → ↓ cerebral perfusion and hypoxia.
2. **Cardiovascular Symptoms:**
 - **Hypotension** → ↓ circulatory volume.
 - **Tachycardia** with weak, **thready pulse** → ↓ blood flow.
3. **Skin Changes:**
 - **Cool, clammy skin** → vasoconstriction.
 - **Cold, mottled extremities** → poor perfusion.
 - **Pallor**.
4. **Respiratory Symptoms:**
 - **Rapid, shallow breathing (tachypnea)** → acidosis.
5. **Other Symptoms:**
 - **Hypothermia** → ↓ perfusion, sweat evaporation.

- **Thirst and dry mouth** → fluid depletion.
- **Fatigue** → inadequate oxygenation.
- **Fainting** → ↓ blood supply.
- **Oliguria/anuria** → ↓ renal perfusion, afferent arteriolar vasoconstriction.

Management of Shock

Objective: Increase **cardiac output** and improve **tissue perfusion** in critical areas like the **coronary, cerebral, renal, and mesenteric vascular beds**.

A. Resuscitation

- Maintain **airway ventilation** and provide **oxygenation**.
- Position: **Head low**, face turned to one side.
- If needed, perform **endotracheal intubation**.

B. Control of Hemorrhage

- For **hypovolemic shock**:
 - Use **elevation, compression bandage, or ligation** to control bleeding.
 - For **internal bleeding**, operate early after resuscitation.

C. Extracellular Fluid Replacement

- Start fluid replacement **immediately**.
- Monitor with **CVP (Central Venous Pressure)**.
- Use **normal saline** or **Ringer's lactate** rapidly until BP is restored.
- If blood is unavailable, use substitutes like **plasma** or **dextran**.

D. Correct Acid-Base Disturbance

- Use **sodium bicarbonate** if arterial blood pH is < **6.2**.
- Correct **electrolyte abnormalities** (e.g., **potassium, calcium**).

E. Drugs

1. **Sedatives:**
 - Relieve pain and anxiety.
 - **Morphine (IV)** is preferred; avoid IM due to delayed absorption in shock.
 - **Pethidine** is also effective.
2. **Chronotropic Agents:**
 - For low heart rate: Use **atropine** or **isoproterenol**.
3. **Inotropic Agents:**

- Enhance cardiac muscle contraction (e.g., **dopamine, dobutamine**).
- Benefits:
 - ↑ **Myocardial activity**.
 - ↑ **Renal blood flow** via vasodilation.
- Use carefully in **small doses**.
- 4. **Vasodilators:**
 - For **septic, traumatic, cardiogenic shock**.
 - Examples: **Nitroprusside, nitroglycerin** (short-acting and reversible).
- 5. **Vasoconstrictors:**
 - Used in **neurogenic shock** to:
 - ↑ **BP** and **coronary circulation perfusion**.
 - ↑ **Myocardial activity**.
 - Examples: **Phenylephrine, metaraminol**.
 - **Avoid in hypovolemic and traumatic shock**.
- 6. **Beta-Blockers:**
 - For **stiff myocardium** and **rapid heart rate** in cardiogenic shock.
 - Example: **Propranolol**.
- 7. **Diuretics:**
 - For **cardiogenic shock** to ↓ vascular volume and filling pressure.
 - **Avoid in hypovolemic and septic shock**, as they worsen hypovolemia.

Q3. Discuss the different types of haemorrhage and their management.

Definition of Hemorrhage

Hemorrhage is the escape of blood from blood vessels, usually due to rupture.

Types of Hemorrhage

I. Based on the Type of Blood Vessel:

- **Arterial: Bright red blood** spurts in sync with the pulse.
- **Venous: Dark red blood** flows steadily.
- **Capillary: Bright red blood** oozes slowly.
- **External:** Bleeding through natural openings (e.g., mouth, nose) or skin.
- **Internal:** Bleeding occurs within the body.

II. Based on Timing:

- **Primary:** Occurs immediately at the time of injury or surgery.
- **Reactionary:** Occurs within 4-6 hours post-injury due to **ligature slip, clot dislodgement**, or **vasospasm cessation**.
- **Secondary:** Occurs **7-10 days** post-injury, often due to **infection** and **arterial wall sloughing**.

III. Based on Blood Loss (ATLS Classification):

- **Class I:** Loss of up to **15%** of blood volume.
- **Class II:** Loss of **15-30%** of blood volume.
- **Class III:** Loss of **30-40%** of blood volume.
- **Class IV:** Loss of **>40%** of blood volume.

Management of Hemorrhage

1. General Management (Hospitalization)

- **A, B, C (Airway, Breathing, Circulation):** Ensure **airway is clear** and provide **oxygen**.
- **IV Line:** Use **isotonic saline** or a balanced crystalloid to restore blood volume.
- **Rest and Sedation:** Calm the patient to reduce bleeding from excitement and anxiety.

2. Specific Measures to Stop Hemorrhage

A. Conservative Methods:

- **Pressure:** Apply firm pressure with sterile gauze (or any cloth if sterile isn't available).
- **Tourniquet:** Useful in **limb hemorrhage**; apply on arm or thigh, max 1 to 1.5 hours.
- **Elevation:** Raise the bleeding area to reduce venous bleeding.

B. Operative Methods:

- **Clamping:** Use **catgut**, **thread**, or **silk** to tie off bleeding vessels.
- **Thermocautery/Diathermy:** For cauterizing blood vessels.
- **Adrenaline Swabs:** Local application for vasoconstriction.
- **Silver Clips:** Used in surgeries, especially **neurosurgery**.
- **Needle and Suture:** Suture vessels directly.
- **Muscle Graft:** Apply crushed muscle at the bleeding site.
- **Bone Wax or BIPP Paste:** For **bone bleeding**.
- **Vascular Repair:** Use vein patches or **Dacron mesh**.
- **Organ Excision:** Remove bleeding organs (e.g., **splenectomy**).

3. Restoration of Blood Volume

- **Fluid Replacement:** Use **Ringer's lactate** to prevent **hyperchloremic acidosis**.
 - Avoid **hypotonic dextrose solutions** in hemorrhage.
- **Blood Transfusion:**
 - **Packed Red Blood Cells (PRBCs):** For volume and oxygen capacity.

- **Platelet Transfusion:** For **thrombocytopenia** with platelet counts $< 20,000\text{--}50,000/\text{mm}^3$.

Q4. Describe the pathophysiology, clinical features and treatment of septic shock.

Septic Shock

- A type of **distributive shock** caused by **endotoxins** in the blood, mostly released by **Gram-negative bacteria** like:
 - *Escherichia coli*
 - *Proteus* species
 - *Klebsiella pneumoniae*
- **Gram-positive bacteria** (e.g., *streptococci*, *pneumococci*) and certain **fungi** can also cause it.
- Leads to **widespread vasodilation** and **sepsis-induced hypotension** (SBP < 90 mmHg or $\downarrow 40$ mmHg from baseline) despite fluids.

Pathophysiology

1. **Cause:**
 - ~70% of cases by **endotoxin-producing Gram-negative bacilli**.
 - Endotoxins = bacterial **lipopolysaccharides (LPS)**.
 - Gram-positive toxins and fungal components can also trigger it.
2. **Mechanism:**
 - **LPS binds** to LPS-binding protein $\rightarrow$ attaches to receptor **CD14** on monocytes/macrophages.
 - Activates **Toll-like receptor (TLR-4)** $\rightarrow$ release of **cytokines** like **IL-1** and **TNF**.
 - Cytokines:
 - Cause **vasodilation** $\rightarrow$ **hypotension**.
 - Reduce **anticoagulation factors** (e.g., thrombomodulin).
 - Results:
 - **Endothelial injury** $\rightarrow$ systemic leukocyte adhesion & lung damage.
 - Activation of **coagulation system** $\rightarrow$ **DIC** (Disseminated Intravascular Coagulation).
 - Combined effects (vasodilation, myocardial failure, DIC) $\rightarrow$ **multiorgan failure** (liver, kidney, CNS).
 - Without rapid treatment, leads to **death**.

Clinical Features

- **Fever (Pyrexia)** with **chills** and **rigors** ($\uparrow$ cytokines/infection).
- **Hypotension** due to **vasodilation**.
- **Warm, sweaty skin** from vasodilation.
- **Anxiety, restlessness, altered mental state** ($\downarrow$ cerebral perfusion $\rightarrow$ hypoxia).

- **Hyperventilation** → **respiratory alkalosis** (medullary stimulation by endotoxins).

Treatment of Septic Shock

- **Prompt diagnosis** and **effective management** are crucial.

1. Infection control

- **Antibiotics:** Start immediately to control infection.
- **Surgical debridement** or **radiological drainage:** Remove dead and septic foci.

2. Fluid replacement

- Correct **fluid deficits** using:
 - **Crystalloids:**
 - **0.9% sodium chloride** (normal saline).
 - **Lactated Ringer's solution.**
 - **Colloids:**
 - **Albumin, dextrans, pentastarch.**
- Guide therapy by monitoring **pulmonary capillary wedge pressure (PCWP)** and **cardiac output.**
- Large volumes may be required initially (~4L or more).

3. Vasopressor therapy

- If fluids fail, use **vasopressors** to restore **hemodynamic stability:**
 - **Dopamine, norepinephrine, epinephrine, phenylephrine.**
 - Maintain **blood pressure** and improve **organ perfusion.**

4. Vasoactive drugs

- Improve **cardiac, pulmonary, and renal function.**
 - Examples: **Phenoxylamine, isopropanolol** (dopamine).

5. Mechanical ventilation

- Late-stage shock often needs **endotracheal intubation** and **ventilation** to ensure proper oxygenation.

6. Steroid therapy

- Indicated only for:
 - **Hypoadrenalism.**

- **Stress coverage** (e.g., recent steroid use for immunosuppression).
- Dose: **Methylprednisolone or dexamethasone** (15-30 mg/kg IV), repeat after 4 hours if needed.

**Q5. Describe pathophysiology, clinical features of hemorrhagic shock .
How will you manage a case of primary hemorrhage after a dental extraction.**

Haemorrhagic Shock

- Defined as shock due to **sudden & rapid loss of significant blood volume**.

Pathophysiology

1. **Initial changes:**
 - **Decreased cardiac output (CO)** and **pulse pressure** → detected by **baroreceptors** in aortic arch and atrium.
 - Blood redistributed to vital organs by increasing **peripheral vascular resistance (PVR)**.
 - **Tachycardia** compensates for low stroke volume.
2. **Compensatory mechanisms:**
 - **Sympathetic activity** increases → stronger heart contractions & venous return.
 - **Interstitial fluid** shifts into blood vessels to maintain volume.
 - **Arteriolar constriction** → ↓ renal blood flow.
 - **Epinephrine & norepinephrine** released → **vasoconstriction, tachycardia** → ↑ BP and CO.
 - **ACTH release** stimulates stress response.
 - **ADH release** → ↑ water reabsorption.
 - **Renin-angiotensin system** activates:
 - **Angiotensin II** causes vasoconstriction.
 - **Aldosterone** ↑ sodium & water retention.
3. **Tissue hypoperfusion:**
 - Leads to **anaerobic metabolism** → **acidosis**.
 - **Hypoxia & mediator release** trigger systemic inflammatory response.

Clinical Features

- **Mental status:** Anxiety, restlessness, confusion (↓ cerebral perfusion).
- **Cardiovascular:**
 - **Hypotension** (↓ blood volume).
 - **Tachycardia** with weak pulse.
 - **Cool, clammy skin** due to vasoconstriction.

- **Respiratory:**
 - **Tachypnea** (rapid shallow breathing from acidosis).
- **Other signs:**
 - **Hypothermia** (↓ perfusion).
 - **Thirst, dry mouth** (fluid loss).
 - **Oliguria/anuria** (↓ kidney perfusion).
 - **Cold, mottled skin** (poor perfusion).
 - **Fatigue & fainting** (low oxygen delivery).

Classification

of Haemorrhagic Shock

Class	Blood Loss	% Volume	Pulse	BP	Resp. Rate	Urine Output	Mental State
I	Up to 750 mL	Up to 15%	<100	Normal	14–20	>30 mL/hr	Slightly anxious
II	750–1500 mL	15–30%	>100	Normal	20–30	20–30 mL/hr	Mildly anxious
III	1500–2000 mL	30–40%	>120	Decreased	30–40	5–15 mL/hr	Anxious/confused
IV	>2000 mL	>40%	>140	Decreased	>35	Negligible	Confused/lethargic

Management of Primary Haemorrhage After Dental Extraction

1. **Post-extraction bleeding** usually occurs in **small vessels**.
2. **Clots extending out of the socket** are removed with **gauze**.
3. A **4-inch gauze pad** (folded) or a **tea bag** is placed over the socket.
4. The patient is instructed to **bite with continuous pressure** for **1 hour**.
 - This procedure may need to be repeated **2-3 times**.
5. **Wait at least 1 hour** before checking the site to avoid disrupting clot formation.
6. **Mouthful of saliva with a few drops of blood** can look like more blood than is actually present.

If Bleeding Continues:

1. **Anaesthetize** the site using **nerve block** or **local infiltration** with **2% lidocaine** containing **1:100,000 epinephrine**.
2. **Curette the socket** to remove existing clot and **freshen the bone**.
3. **Irrigate with normal saline**.
4. **Suture under gentle tension**.
5. Place **local haemostatic agents**:
 - **Oxidized cellulose**
 - **Topical thrombin on a gelatin sponge**
 - **Microfibrillar collagen**before suturing.

Q6. Describe neurogenic shock and its management.

Neurogenic Shock

A type of **distributive shock** caused by loss of **sympathetic tone** due to disruption of the **sympathetic nervous system**.

Causes of Neurogenic Shock

- **Spinal cord trauma**
- **Spinal cord neoplasm**
- **Spinal/epidural anesthesia**
- Commonly occurs with **cervical or high thoracic spinal cord injuries**, disrupting sympathetic regulation of vascular tone.

Pathophysiology

1. **Loss of sympathetic tone** leads to:
 - **Massive vasodilatation**
 - Inhibition of **baroreceptor response**
 - Impaired **thermoregulation**
2. **Arterial vasodilatation** → **hypotension** → **reduced cardiac output** → **impaired tissue perfusion**.
3. Vascular compromise to the spinal cord:
 - **Loss of autoregulation**
 - **Vasospasm and thrombosis**
4. **Neurotransmitter accumulation** and release of **free radicals**.
 - Hypotension worsens **acute spinal cord injury** by reducing blood flow to the spinal cord.

5. Disruption of **sympathetic input to the heart** (normally increases heart rate and contractility), leading to absence of **reflex tachycardia** during hypovolemia.

Clinical Features

- History of **emotional stress or sudden pain**.
- **Hypotension with bradycardia** (no reflex tachycardia).
- **Warm, dry extremities** due to loss of **peripheral vasoconstriction**.
- **Motor and sensory deficits** (spinal cord injury).
- **Tachypnea, unconsciousness, anxiety, restlessness**.
- **Reflexes usually intact**.

Management of Neurogenic Shock

1. **Airway potency** and adequate **ventilation**.
2. **Fluid resuscitation** to restore **intravascular volume**:
 - **Most patients respond** to fluid restoration, improving perfusion and resolving hypotension.
3. **Vasoconstrictors** (dopamine, phenylephrine):
 - **Improve vascular tone, decrease capacitance, increase venous return**.
 - Only after excluding **hypovolemia** as the cause of hypotension.
 - The duration of **vasopressor support** may affect the **prognosis** and improvement in **neurologic function**.
4. Rapid restoration of **blood pressure** and **circulatory perfusion**:

Improves perfusion to the spinal cord, preventing **ischemia** and minimizing **secondary cord injury**

Q7. Describe indications and complications of blood transfusion.

Definition of Blood Transfusion

Blood transfusion is the process of transferring blood or blood-based products from one person to another.

Types of Blood Transfusions:

- **Homologous (Allogeneic)**: Uses **blood from a donor**.
- **Autologous**: Uses **the patient's own stored blood**.

Indications of Blood Transfusion

- **Acute Hemorrhage**: **Internal or external bleeding**.
- **Major Operations**: Where **significant blood loss** is expected.

- **Deep Burns:** Due to **tissue damage** and **red blood cell (RBC) destruction**.
- **Severe Anemia:** **Hemoglobin < 10 gms%**, especially pre- or post-surgery.
- **Blood Disorders:** Conditions like **hemophilia, leukemia, thalassemia**.
- **Erythroblastosis Fetalis:** Due to **Rh incompatibility** in newborns; use **exchange transfusion**.
- **Chemotherapy for Cancer:** May **suppress blood formation**.
- **Debilitating Diseases:** Such as **uremia, liver failure, and hypertension**.

Complications of Blood Transfusion

A. Transfusion Reactions

I. Incompatibility

Reasons for incompatibility:

- **Mistake in crossmatching:** Errors due to old sera or wrong labeling.
- **Hemolysis of blood:** Blood may get destroyed due to over-freezing, shaking, or warming.
- **Expired blood:** Using blood after its expiry date.

Clinical Features of Mismatched Blood Transfusion:

- **Rigors and fever**
- **Nausea, vomiting, headache**
- **Pain in the loins** (due to acute tubular necrosis)
- **Chest pain and dyspnoea**
- **Loss of consciousness**
- **Reduced urine output and haemoglobinuria** (within 2-3 hours)
- **Jaundice** after 24-36 hours (confirmatory evidence)
- **Renal failure** (due to blockage of renal tubules by haematin pigment)

Treatment:

- **Stop transfusion immediately.**
- **Send blood and urine samples** to the lab for rechecking.
- **Start IV fluids** instead of blood.
- **Alkalize the blood** using sodium lactate and sodium bicarbonate.
- **Frusemide 80-120 mg IV** for forced diuresis (repeat if urine output >30 ml/hr).
- **Give antihistamines and hydrocortisones.**
- **Hemodialysis** if severe.

II. Pyrexial Reactions

Common symptoms:

- **Pyrexia, chills, rigors, headache, tachycardia, nausea, vomiting.**

Causes:

- **Improperly sterilized drip sets.**
- **Pyrogens** in donor blood.
- **Infected blood transfusion.**
- **Rapid transfusion.**

Prevention:

- Use **sterilized plastic sets.**

Treatment:

- **Stop transfusion temporarily.**
- **Antihistamines** and **antipyretics.**
- Resume transfusion after the patient stabilizes.

III. Allergic Reactions

Symptoms:

- **Urticaria, tachycardia, fever, dyspnoea.**
- **Severe anaphylactic shock** (rare).

Treatment:

- **Stop transfusion.**
- **Antihistamines** and **cortisones.**
- **Treat shock.**

B. Transmission of Diseases

1. **Serum Hepatitis: Hepatitis B** can be transmitted through blood transfusion. Symptoms appear within **3 months.**
2. **AIDS: HIV** transmission via donor blood.
3. **Bacterial Infection:** Caused by **faulty storage** of blood, treated with **antibiotics** to prevent sepsis.

C. Reaction Due to Massive Blood Transfusion

Massive blood transfusion: 8-10 units of blood in 24 hours.

Complications:

- **Acid-base imbalance** causing metabolic alkalosis.
- **Hyperkalaemia.**

- **Citrate toxicity:** Consumes calcium, causing **hypocalcaemia**. Calcium may need to be infused.
- **Hypothermia:** Blood from the fridge causes **temperature drop (3-4°C)**.
- **Failure of coagulation:** Dilution of platelets and clotting factors.
- **DIC** (Disseminated Intravascular Coagulation) after mismatched transfusion.

D. Complications of Over Transfusion

- Seen in **chronic anaemia, children, and elderly patients**.
- **Packed cells** should be given instead of whole blood.
- **Slow transfusion** over **4-6 hours** with **intervals**.
- **Elderly patients** should receive packed cells with **diuretics**.

E. Complications of Intravenous Transfusions

- **Thrombophlebitis**.
- **Air embolism**.

F. Pulmonary Complications

- **TRALI** (Transfusion-Related Acute Lung Injury): Non-cardiogenic pulmonary oedema caused by transfusion.

Q8. Discuss the Management of Hypovolaemic Shock.

1. Resuscitation

- Maintain **airway ventilation and oxygenation**.

2. Position

- **Head low**, face turned to one side.
- **Intubation if necessary**.

3. Control Hemorrhage

4. Fluid Resuscitation:

- Start **immediately** via a large **gauge needle/cannula**.
- Monitor **CVP** to assess fluid replacement.

Fluids Used:

- **Lactated Ringer's Solution:** Restores **extracellular fluid deficit**.
- **Hypertonic Saline:** Effective for initial resuscitation.
- **Hetastarch:** Artificial colloid, cheaper than albumin.
- **Dextran:** Used as **plasma expander**.
- **Blood substitutes:** Plasma/dextran if blood is unavailable.
 - **DIC (Disseminated Intravascular Coagulation):** Especially after **mismatched blood**.

Q9. Triage trauma

Triage in Trauma

(General Surgery, BDS 3rd Year)

Definition

- **Triage** is the process of **sorting patients** based on the severity of their injuries to provide **immediate treatment** in emergency situations, especially during **mass casualty incidents**.
- The goal is to prioritize patients who need **urgent care** to maximize survival.

Levels of Triage

1. **Red (Immediate):**
 - **Life-threatening injuries** requiring immediate intervention.
 - Examples: **Severe bleeding, respiratory distress, cardiac arrest**.
2. **Yellow (Delayed):**
 - **Serious injuries** but can wait for treatment without immediate risk to life.
 - Examples: **Fractures, burns** that are not life-threatening.
3. **Green (Minor):**
 - **Minor injuries** that do not require urgent care.
 - Examples: **Superficial cuts, abrasions, and sprains**.
4. **Black (Deceased or Expectant):**
 - **Non-survivable injuries** or patients who are **clinically dead**.
 - Focus is on those with better chances of survival.

Principles of Triage

- **Rapid Assessment:** Patients are quickly assessed to determine the **severity of injuries**.

- **Maximizing Survival:** Prioritize those who have the **best chance of survival** with available resources.
- **Minimizing Delays:** Treatment is **streamlined** to prevent unnecessary delays in care.
- **Resource Allocation:** In large-scale emergencies, resources (medications, staff, equipment) are used where they are most needed.

Methods of Triage

1. **START (Simple Triage and Rapid Treatment):**
 - Used in **disasters** or **mass casualty events**.
 - Involves a **quick check of vital signs** (breathing, pulse, mental status) to assign priority levels.
2. **SALT (Sort, Assess, Life-saving interventions, Treatment/Transport):**
 - **Similar to START**, but includes additional steps for **life-saving interventions**.
3. **MIMMS (Major Incident Medical Management and Support):**
 - Used for **organized triage** in large-scale incidents, with a focus on **patient tracking** and **team coordination**.

Triage in Dental Trauma

- In **dental trauma**, triage focuses on prioritizing **airway management**, control of **bleeding**, and evaluation of **fractures**.
- **Severe injuries** like **avulsion** or **fractured jaw** should be treated immediately, while **minor fractures** can wait.

Q10. Causes, treatment and management of Vasovagal Shock

Causes of Vasovagal Shock

- **Emotional stress:** Intense feelings like fear, pain, or anxiety can trigger a vasovagal response.
- **Prolonged standing:** Staying in one position for a long time, especially in hot conditions, can cause a sudden drop in blood pressure.
- **Pain:** Severe pain, especially after surgery or injury, can trigger vasovagal syncope.
- **Vagal nerve stimulation:** Overstimulation of the vagus nerve, leading to a drop in heart rate and blood pressure.
- **Sight of blood or injury:** Sometimes, seeing blood or injury can cause a vasovagal response, leading to fainting.

Treatment of Vasovagal Shock

- **Trendelenburg position:** The patient should be laid flat with their legs elevated to improve blood flow to the brain.
- **Fluids:** Administering intravenous fluids (IV) to increase blood volume and improve circulation.
- **Oxygen:** Providing oxygen to support breathing and circulation, especially if the patient is hypoxic (low oxygen levels).
- **Medication:** In severe cases, atropine may be given to increase heart rate and improve circulation.

Management of Vasovagal Shock

- **Monitor vital signs:** Regularly check the heart rate, blood pressure, and oxygen levels to track the patient's condition.
- **Identify triggers:** Identifying and avoiding triggers, such as stress or standing for long periods, is key in preventing recurrence.
- **Educate the patient:** Teach the patient techniques to prevent fainting episodes, such as lying down when feeling faint or avoiding sudden changes in posture.
- **Reassurance:** Reassure the patient, as anxiety and fear can worsen symptoms.

Key Points

- Vasovagal shock is a temporary condition caused by sudden drops in heart rate and blood pressure.
- Immediate treatment focuses on improving blood circulation and oxygen supply to the brain.
- Long-term management involves identifying triggers and patient education to prevent recurrence.